

Multiple Choice

1. **Answer:** E

Options: A) Through ultrasound examination; B) Through blood sample analysis; C) Through direct observation; D) Through measuring brain wave activity in response to sounds; E) Through operant conditioning

Note: Operant conditioning allows researchers to assess a dog's hearing sensitivity by reinforcing specific responses to sounds of varying frequencies, thus enabling the measurement of their auditory discrimination capabilities.

2. **Answer:** A

Options: A) The controls are nerve impulses in the body; B) The controls are the contraction and relaxation cycles of skeletal muscles; C) The controls are hormonal responses that regulate digestion and absorption of food; D) The controls are mechanisms that regulate the heart rate; E) The controls are processes that control the body temperature

Note: The regulation of plasma and extracellular fluid volume involves nerve impulses that activate mechanisms such as the autonomic nervous system and hormonal responses, which work together to maintain homeostasis in the body.

3. **Answer:** C

Options: A) Appetitive behavior, exploratory behavior, quiescence; B) Termination, appetitive behavior, exploratory behavior; C) Appetitive behavior, consummatory act, quiescence; D) Consummatory act, exploratory behavior, termination; E) Appetitive behavior, termination, consummatory act

Note: The correct answer identifies the stages of a behavioral act as appetitive behavior, which involves the search for a goal, the consummatory act, which is the actual achievement of that goal, and quiescence, which is the resting state following the completion of the behavior, reflecting the sequential nature of behavior in biological processes.

4. **Answer:** C

Options: A) opt for zero population control once the K value of the curve has been reached; B) maintain the population at the highest point of its logistic curve; C) reduce the carrying capacity of the environment to lower the K value; D) decrease the mortality rate; E) increase the birth rate of the species

Note: Reducing the carrying capacity of the environment lowers the K value, which decreases the maximum sustainable population size and effectively controls the mosquito population by limiting resources necessary for their growth and survival.

5. **Answer:** D

Options: A) Coevolution; B) Punctuated equilibrium; C) Directional selection; D) Divergent evolution; E) Convergent evolution

Note: Divergent evolution occurs when two related species develop different traits and

adaptations over time, leading to increased differences between them as they adapt to distinct environments or ecological niches.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The short- and long-term outcomes after LRT and SLT did not differ significantly. To avoid the risk for the donor in LRT, SLT represents the first-line therapy in pediatric liver transplantation in countries where cadaveric organs are available. LRT provides a solution for urgent cases in which a cadaveric graft cannot be found in time or if the choice of the optimal time point for transplantation is vital.

7. **Answer:** True

Options: A) True; B) False

Note: The available evidence suggests that the operative treatment for DMCF is associated with a lower rate of nonunion, malunion and complication than nonoperative treatment. This study supports traditional primary operative treatment for DMCF in active adults.

8. **Answer:** True

Options: A) True; B) False

Note: For some patients, using colour to describe their pain experience may be a useful tool to improve doctor-patient communication.

9. **Answer:** True

Options: A) True; B) False

Note: Elective re-siting of intravenous cannulae every 48 hours results in a significant reduction in the incidence and severity of PVT. We recommend that this should be adopted as standard practice in managing all patients who require prolonged intravenous therapy.

10. **Answer:** True

Options: A) True; B) False

Note: When there is clinical suspicion of sepsis, appropriate empirical systemic antibiotic therapy should be broad spectrum and should rely on the susceptibility of the organisms from recent cultures of the burn wound surface, until the blood cultures results are completed.

Long Form Response

11. **Answer:** Hormones such as cortisol, aldosterone, and adrenal androgens play significant roles in the physiological changes that occur during puberty in both males and females. Cortisol, a glucocorticoid, is involved in energy metabolism and the stress response, influencing growth and development during this critical period. Aldosterone,

a mineralocorticoid, helps regulate fluid and electrolyte balance, which is essential for the physical changes associated with puberty, such as increased blood volume and body composition alterations. Adrenal androgens, including dehydroepiandrosterone (DHEA), are precursors to sex steroids and contribute to the development of secondary sexual characteristics, such as body hair and changes in fat distribution. Collectively, these hormones orchestrate the complex interplay of metabolic, developmental, and physiological changes that characterize puberty, highlighting their critical roles in sexual maturation and overall growth.

Note: correct because it identifies the specific roles of cortisol in energy metabolism and stress response, aldosterone in fluid and electrolyte balance, and emphasizes their overall contributions to the physical changes that occur during puberty in both sexes.

12. **Answer:** If messenger RNA (mRNA) molecules were not degraded after polypeptide synthesis, the consequences on cellular function and gene regulation would be profound. The accumulation of mRNA would lead to rapid replication and an increase in the expression levels of corresponding proteins, potentially overwhelming cellular machinery. This could disrupt the delicate balance of gene expression, leading to aberrant protein levels that might impair cellular homeostasis and function. Furthermore, the lack of mRNA degradation would hinder the ability of the cell to respond to changing environmental conditions, as it would be unable to efficiently regulate gene expression in response to signals. Ultimately, this dysregulation could contribute to pathological states, including cancer, where uncontrolled protein synthesis is a hallmark.

Note: The answer correctly identifies that the failure to degrade mRNA post-synthesis would result in excessive protein production, disrupting gene expression balance and potentially harming cellular function and homeostasis.